Static analysis and software verification

Lecture 7 - Introduction to abstract interpretation (Galois Connections)

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Concrete domain

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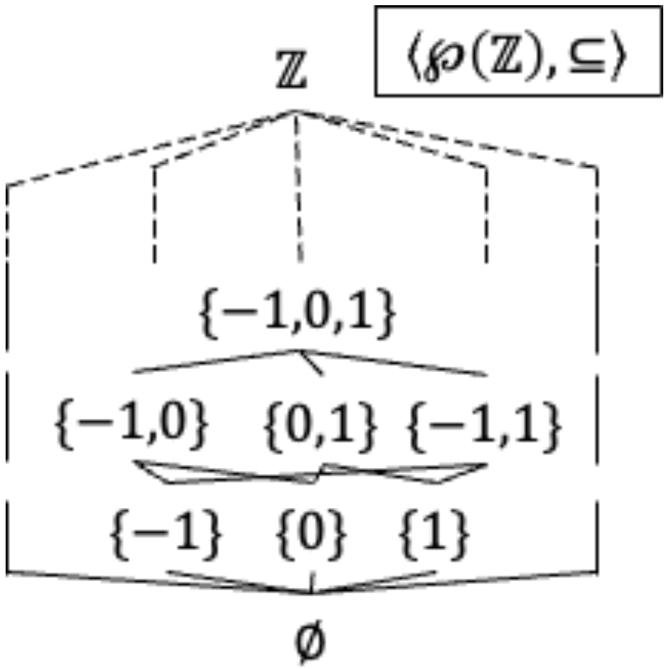
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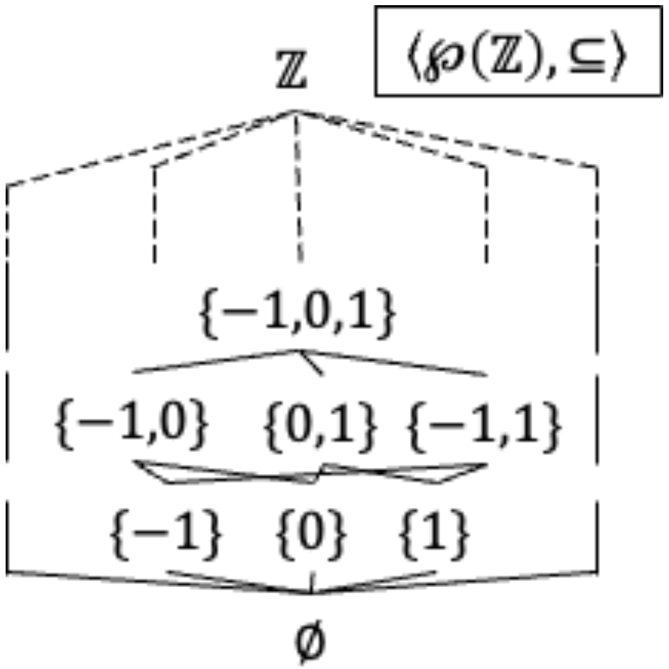
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- Join point: union
- $b_1 \wedge b_2$: intersection
- input of the user: top
- unreachable point: bottom

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  - Concretization function: $\gamma : A \to C$

Constant propagation

From previous lectures

"For each program point, whether or not a variable has a constant value whenever an execution reaches that point"

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  - Sound iff the result is $\supseteq$

Sign abstract domain

Example

Soundness

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The other direction

- Sound abstract domain (via abstraction and concretization)
- Intuition
  - Start from an abstract value
  - Concretize it (via γ)
  - Go back to the abstract (via α)
  - Sound iff the result is $\sqsupseteq_A$ (in general, =)

Sign abstract domain

Example

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  - If we concretize something that represents less information than something else, we obtain something with less or the same information

Sign abstract domain

Example

Galois Connection (GC)

- Let $\langle C, \sqsubseteq_C \rangle$ and $\langle A, \sqsubseteq_A \rangle$ two posets, $\alpha : C \to A$ and $\gamma : A \to C$ two monotone functions. They form a Galois Connection if

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- $c \sqsubseteq_C \gamma(a) \implies \alpha(c) \sqsubseteq_A a$: this ensures the **best abstraction**

Galois Connection (GC)

Alternative characterization of GC

- Let $\langle C, \sqsubseteq_C \rangle$ and $\langle A, \sqsubseteq_A \rangle$ two posets, $\alpha : C \to A$ and $\gamma : A \to C$, we have a GC if α and γ satisfies all the following properties
  - α and γ are monotone
  - $\gamma \circ \alpha$ is extensive, i.e., $\forall c \in C \,.\, c \sqsubseteq_C \gamma(\alpha(c))$
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    - Often =

Constant propagation

Proofs

- α and γ are monotone
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  - In this case the abstraction is surjective and the concretization is injective
- Every GC can be reduced to a GI!
- When we do not have a GI, we have at least two abstract elements that represent the same information (and thus the redundant elements can be removed by domain reduction)

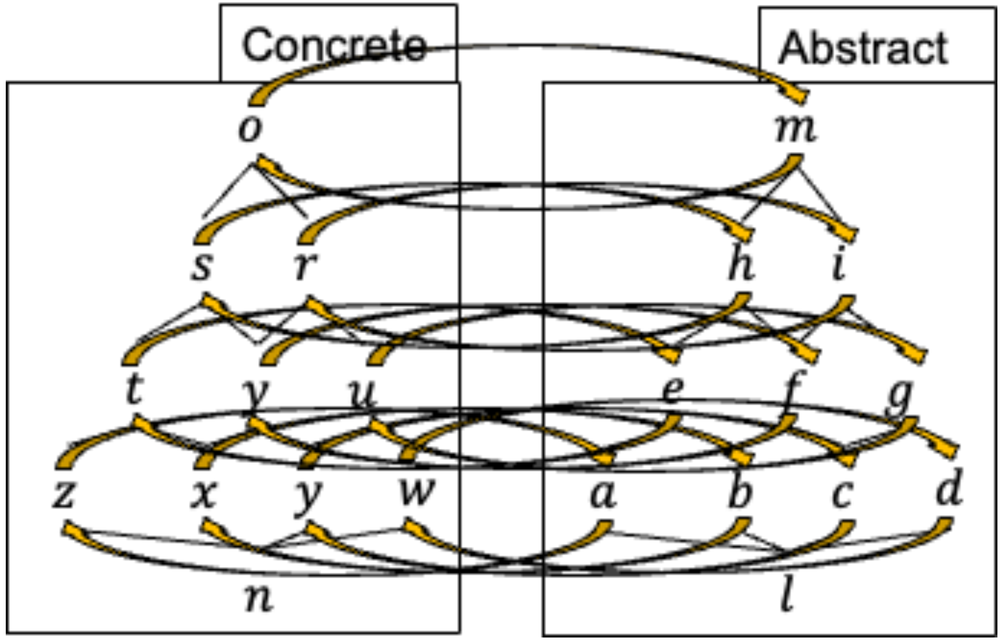
Constant propagation

Formalization

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$\gamma \circ \alpha = id?$

- $\forall v \in C \,.\, \gamma(\alpha(v)) = v$
  - (almost) never true
- It means there is no abstraction (and the abstract domain is a isomorphism)
- and the computation on concrete and abstract is (presumably) the same
- NOT what we want



On the best abstraction

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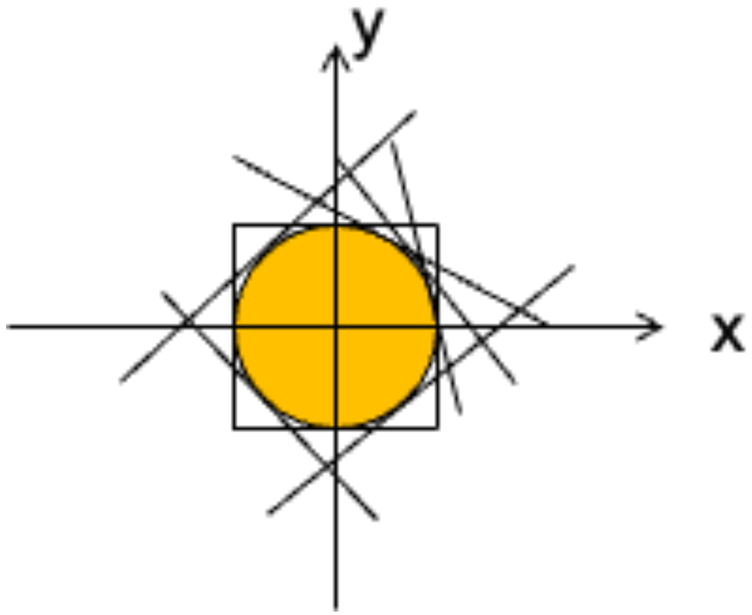
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- It may happen that the GC between C and A does not exist, due to the non-existence of the best abstraction (e.g., convex polyhedra)

On the best abstraction

Example

- $P = \{(x, y) \mid x^2 + y^2 \leq 1\}$
- With a convex polyhedron:
  $a * x + b * y \leq c$
- but we can always find a better approximation



On the minimal abstraction

Example with finite state automata

State of constant propagation

- For now, we just talked about abstraction on values
- But the state of the analysis is $\mathbb{V} \mapsto CP$
  - Pointwise application of CP
- We proved that CP form a GC with $\wp(\mathbb{Z})$, but what about states?
- We can prove that given an abstraction of values forming a GC, its pointwise application is a GC as well

The sign abstract domain

Exercise